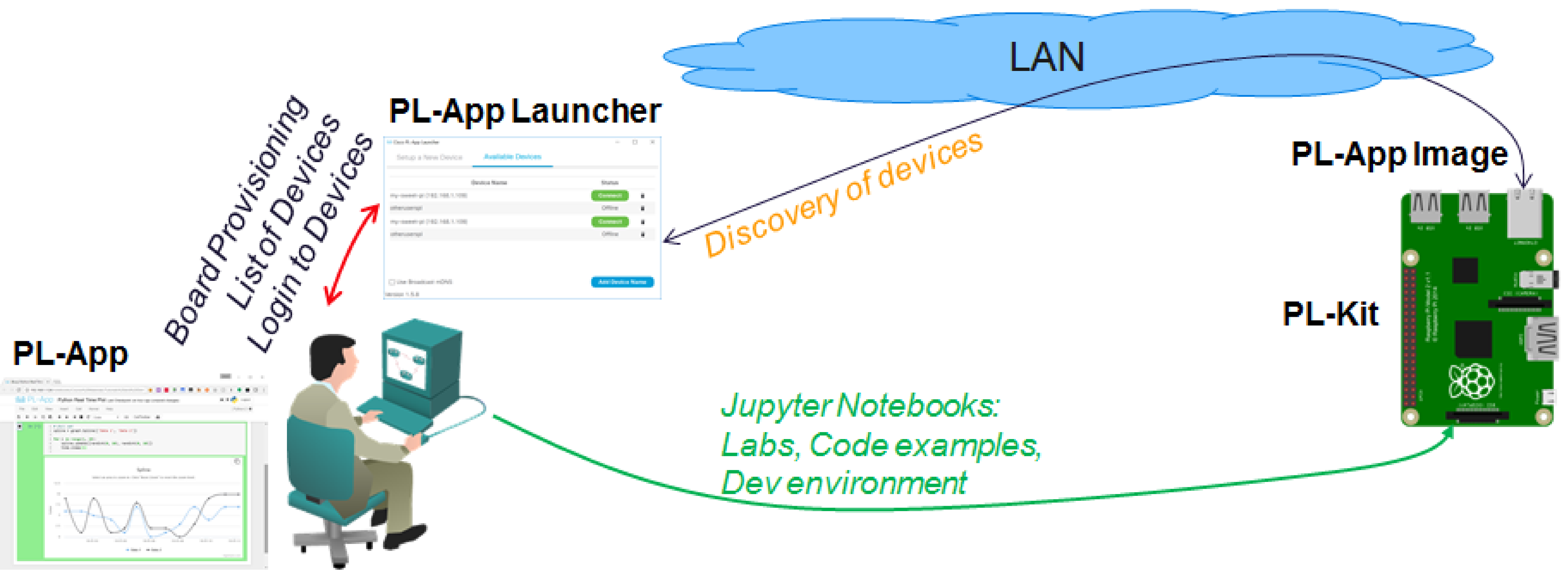
Cisco Networking Academy® Mind Wide Open™

Lab: Writing Scripts Using a Text Editor ¶

Topology



Objectives

- Access PL-App from your Netacad.com class.
- Connect to the Shell on the board using the PL-App Web Shell
- Create a Program Using Nano

Background / Scenario

Writing scripts using a command line editor is a skill that is often used when managing servers and remote devices. This lab introduces the Nano text editor which is available through the BASH shell.

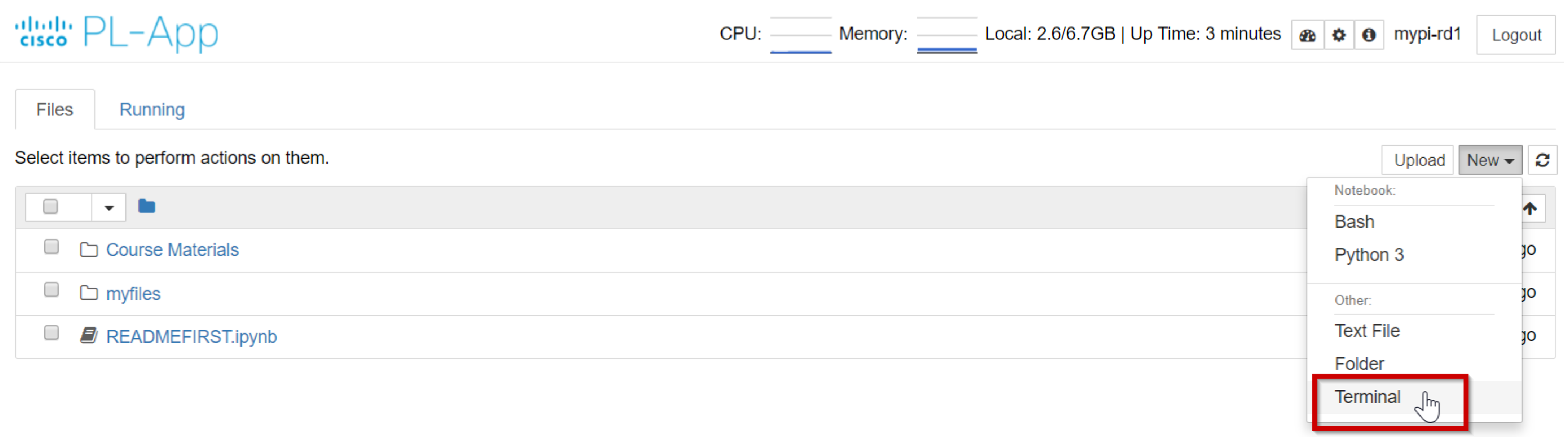
Required Resources

- PC with Internet Access
- Ethernet based connection to the Internet with DHCP and no traffic filtering
- Raspberry Pi that is configured and imaged for PL-App access

Part 1: Accessing the Shell Interface using PL-App

Step 1: Connect to the Linux Shell using the PL-App Web Shell.

a. Click the New menu in PL-App from the menu and select Terminal:



b. A new window will open with the Linux shell running on the Raspberry Pi board.

Step 2: Create a Program Using Nano

1. Open the text editor Nano by typing `nano GuessAgain.py` in the command line. If the `GuessAgain.py` file already exists in the current directory, nano will open the file for editing. If the file does not exist, it will create it. *Note: Filenames are case sensitive in Linux, so make sure you type in the same case each time.*
2. To ensure that the script will be executed by the Python (version 3) interpreter, the very first line of the script needs to start with the shebang `#!` characters followed by the *path/location* of the interpreter (*the env util selects the Python executable that is active in the current $PATH*):

```
#!/usr/bin/env python3
```

3. Enter the remaining code into the `GuessAgain.py` file. Once complete, save the file and exit the editor by pressing CTRL+x and selecting Y, for yes, to save the file then Enter to accept the current filename.

```
#!/usr/bin/env python3
C="X"           #C represents Correct
G=1             #G represents Guesses
L=0             #L represents Low
H=1000          #H represents High
print ("Think of a number between 1 and 1000 and I will guess it.")
while C != "C":
    print ("Guess Number ",G)
    print ("Is your number ",int((L+H)/2),"? (C)orrect, (L)ower, (H)igher") #make sure it is integer
    C = input(">")
    if C == "L":
            H = (L+H)/2
            G = G + 1
    else:
            if C == "H":
                    L = (L+H)/2
                    G = G + 1
            else:
                    print ("Please enter C,L, or H")
print ("I guessed it in ",G,"tries!")
```

4. Before you can directly execute the Python code from the shell, you must turn the file into an executable file using the `chmod +x` command.

   Execute the program by typing `./GuessAgain.py`.

```
root@chestnut:/home/pi/notebooks# ls -l ./GuessAgain.py
-rw-r--r-- 1 root root 1689 Dec  5 02:24 GuessAgain.py
root@chestnut:/home/pi/notebooks# ./GuessAgain.py
bash: ./GuessAgain.py: Permission denied
root@chestnut:/home/pi/notebooks#  chmod +x GuessAgain.py
root@chestnut:/home/pi/notebooks# ls -l ./GuessAgain.py
-rwxr-xr-x 1 root root 1689 Dec  5 02:24 GuessAgain.py
root@chestnut:/home/pi/notebooks#  ./GuessAgain.py
```

5. Examine the code from the command line using the cat, more, and less commands.

```
root@chestnut:/home/pi/notebooks#  cat GuessAgain.py
root@chestnut:/home/pi/notebooks#  more GuessAgain.py
root@chestnut:/home/pi/notebooks#  less GuessAgain.py
```

Note: Press Q to quit the less command.

Reflection

In the GuessAgain.py program, when C is pressed for correct, the program asks the user to Please enter C,L, or H before quitting. Can you fix this problem?